

Hedging European Options under Alternative Volatility Models

Based on Historical Backtest using Black–Scholes, GARCH, and CEV

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Course Project Presentation

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Project Description and Goal

This project studies the historical performance of **delta hedging** for European call options under three volatility specifications:

- **Black–Scholes (BS)**: constant volatility benchmark,
- **GARCH-BS**: rolling time-varying volatility forecast plugged into Black–Scholes,
- **CEV**: state-dependent volatility model with Monte Carlo pricing and numerical delta.

The goal is to compare these models under the same rolling backtest and determine which model gives the smallest hedging errors in practice.

For each rolling start date, the procedure is:

- 1 calibrate the model on a historical lookback window,
- 2 price one European option,
- 3 compute the model-implied delta,
- 4 rebalance daily until maturity,
- 5 compare terminal hedge value against option payoff.

Mathematical Setup

For each trade date:

$$S_0 = \text{initial stock price}, \quad K = mS_0, \quad \tau_0 = \frac{H}{252},$$

where m is moneyness and H is maturity in trading days.

The self-financing hedge portfolio is

$$\Pi_t = \Delta_t S_t + B_t, \quad B_0 = V_0 - \Delta_0 S_0.$$

At each daily rebalance:

$$B_j = B_{j-1} e^{r/252} - (\Delta_j - \Delta_{j-1}) S_j.$$

At maturity:

$$\Pi_T = \Delta_T S_T + B_T, \quad \text{Hedge Error} = \Pi_T - \Phi(S_T).$$

We compare models using

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |e_i|, \quad \text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N e_i^2}.$$

Model Formulas

Black–Scholes benchmark:

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t, \quad \Delta^{\text{BS}} = e^{-qT} \Phi(d_1).$$

GARCH(1,1):

$$\begin{aligned} r_t &= \mu + \varepsilon_t, & \varepsilon_t &= \sigma_t Z_t, \\ \sigma_t^2 &= \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \end{aligned}$$

CEV model:

$$dS_t = (r - q)S_t dt + \delta S_t^\beta dW_t,$$

so the instantaneous return volatility is

$$\sigma_{\text{return}}(S_t) = \delta S_t^{\beta-1}.$$

Interpretation:

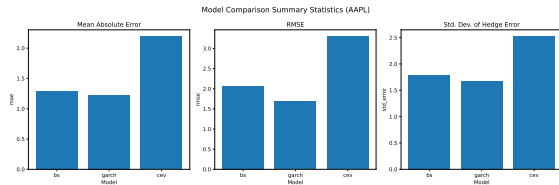
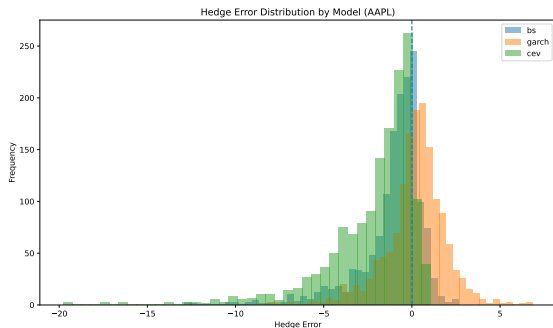
- BS assumes constant volatility.
- GARCH captures volatility clustering.
- CEV captures volatility depending on stock level.

Summary Table: AAPL, AMZN, META

Ticker	Model	MAE	RMSE	Mean Error	Std. Error
AAPL	BS	1.2907	2.0666	-1.0240	1.7957
AAPL	GARCH	1.2316	1.6824	0.2032	1.6706
AAPL	CEV	2.1928	3.2971	-2.1262	2.5208
AMZN	BS	1.8615	2.7397	-1.7096	2.1416
AMZN	GARCH	1.6361	2.1914	0.5465	2.1229
AMZN	CEV	2.7261	4.1329	-2.6000	3.2137
META	BS	5.2965	8.6153	-5.2371	6.8431
META	GARCH	5.3249	8.3557	2.2482	8.0503
META	CEV	7.8032	11.9173	-7.7600	9.0476

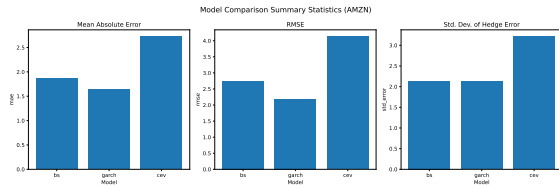
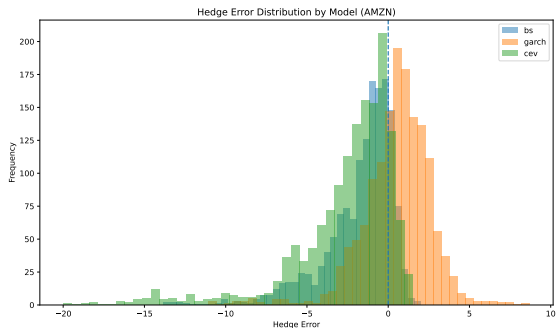
Across all three tickers, CEV is the weakest model by both MAE and RMSE. GARCH gives the lowest RMSE for AAPL, AMZN, and META, while BS is marginally best in MAE for META.

AAPL: Key Graphs



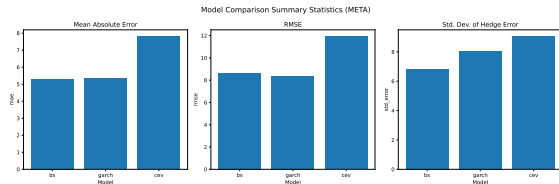
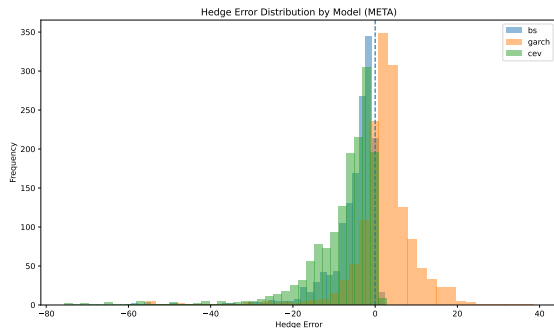
- GARCH is centered closest to zero and has the smallest RMSE.
- CEV shows a noticeably heavier negative tail.

AMZN: Key Graphs



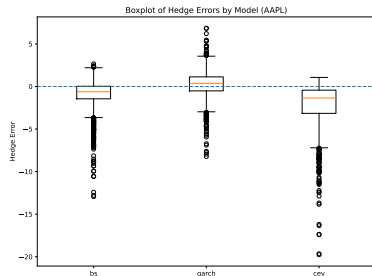
- GARCH dominates by MAE and RMSE.
- CEV again produces the widest downside tail and largest dispersion.

META: Key Graphs

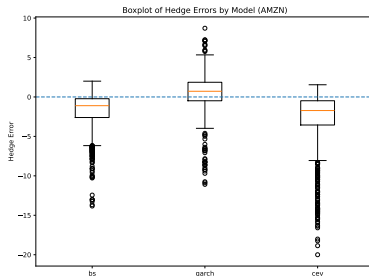


- GARCH gives the best RMSE, while BS is slightly better in MAE.
- CEV is clearly dominated in both average and tail error behavior.

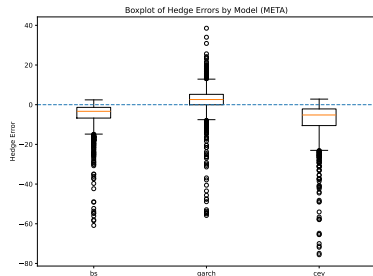
Cross-Ticker Visual Comparison



AAPL



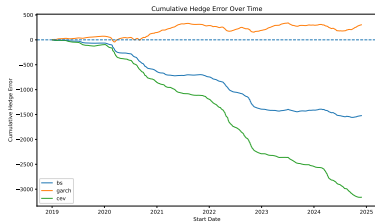
AMZN



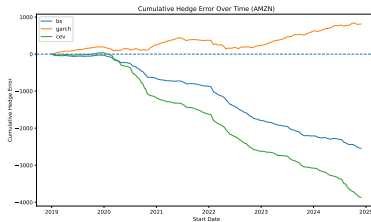
META

- For all three tickers, GARCH has the box closest to zero.
- CEV has the most negative outliers and the weakest tail behavior.

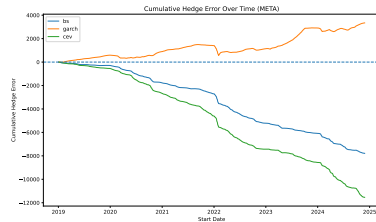
Cumulative Hedge Error over Time



AAPL



AMZN



META

- GARCH remains the most stable over time.
- BS and especially CEV accumulate larger negative hedging error.

Main Takeaways

- **GARCH performs best overall** in this study, especially in RMSE.
- This suggests that **volatility clustering** is the dominant feature for AAPL, AMZN, and META.
- **CEV underperforms** because state-dependent volatility alone does not explain the observed hedge dynamics as well as time-varying volatility persistence.
- **BS remains a useful baseline**, but it is generally less accurate than GARCH.

- Include **transaction costs** and bid–ask spread.
- Compare against **EGARCH**, **GJR-GARCH**, or local volatility models.
- Improve the ML volatility forecaster and add cross-validation.
- Study whether certain sectors or stock classes systematically favor CEV.
- Extend the analysis to **portfolio-level hedging** rather than one option at a time.

Conclusion

This project shows that alternative volatility assumptions matter materially for delta hedging performance. In the uploaded empirical results, GARCH gives the strongest and most stable hedging results for AAPL, AMZN, and META, while CEV performs worst across all three names.

Thank you.